Necessary conditions on minimal-time solutions of the firing squad problem

Kia Keng Giam[a,*]

[a] *Temasek Junior College, Singapore*

Abstract

We prove necessary conditions on the space-time diagrams of all minimal-time solutions of the firing squad synchronization problem, whatever their number of states: the non-periodic part of the half-line reaches the line $i = t/3$, the half-line is not periodic on any wedge at the left border and never eventually regular, and the first rows after the reflection take finitely many forms, determined by a periodicity certificate. With computer assistance, including LRAT-certified unsatisfiability proofs, we show that four states are excluded by the lines of lengths $2, \ldots, 5$ together with the first five rows, the return chains and the right ends of the lines of lengths $6, \ldots, 10$; that five five-state rules synchronize exactly the lines of lengths $2, \ldots, 14$; that every five-state minimal-time solution uses at least 15 neighbourhoods on its half-line below the anti-diagonal $t + i = 78$; and that a conditional five-state result claimed by Balzer in 1967 holds.

Keywords: firing squad synchronization problem, cellular automata, state complexity, minimal time, pumping lemma, proof certificates

1. Introduction

The firing squad synchronization problem, posed by Myhill in 1957 and popularised by Moore [1], asks for a finite automaton that, placed in every cell of a line of arbitrary length n and started with all cells quiescent except a *general* at one end, drives all cells into a firing state for the first time at the same instant. Goto [2] and Waksman [3] solved it in the optimal time $2n - 2$. The least number of states of a minimal-time solution is a benchmark question [4]: Waksman used 16 states, Balzer 8 [5], Gerken 7 [6] and Mazoyer 6 [7]. Balzer's computer search excluded four states [5]; Yunès [8] observed that the lines of lengths $2, \ldots, 9$ suffice, and Sanders [9] confirmed the result with a corrected search. Whether five states suffice has been open since 1987. As there are finitely many k-state rules, k-state minimal-time solutions do not exist if and only if for some N no k-state rule synchronizes all lines of lengths $2, \ldots, N$ in minimal time, so a negative answer may consist of finitely many necessary conditions.

Until the echo of its right border returns, the line of length n behaves like the infinite half-line, so a solution consists of one half-line diagram C_∞ and, for each n, a triangle R_n that is a function of two anti-diagonals of C_∞ (Lemma 3.1). We derive conditions at the right end of R_n (Lemmas 3.3 and 3.5 and Theorem 3.4), at its left end (Lemma 4.1, Theorem 4.2, and Corollary 4.3) and in its first rows (Corollary 5.3 and Lemma 5.5), and apply them to four and five states in Section 6.

Related work. Indistinguishability arguments, in which a solution cannot tell two situations apart and so fires too early, exclude minimal-time solutions with one-way channels [10, Prop. 2] and, through repetitions forced by the pigeonhole principle, for variants with sub-generals or some geometric configurations [11, 12]; our pumping lemmas use this mechanism to constrain solutions of the original problem. The eventual periodicity behind the front is classical [13, 14]; lower bounds for restricted classes are surveyed in [4]. We are not aware of statements equivalent to Theorems 3.4 and 4.2 and Corollaries 4.3 and 5.3 in the literature.

*Corresponding author.
Email address: giamkiakeng@gmail.com (Kia Keng Giam)

2. Lines and the half-line

Let S be a finite set of *states* containing three pairwise distinct states, the quiescent state L, the general G and the firing state F; the *working* states are $W := S \setminus \{\mathsf{F}\}$, and $* \notin S$ denotes the border. A *rule* is a map $\delta : (W \cup \{*\}) \times W \times (W \cup \{*\}) \to S$ with

$$\delta(\mathsf{L},\mathsf{L},\mathsf{L}) = \delta(\mathsf{L},\mathsf{L},*) = \mathsf{L}. \tag{1}$$

The *line of length* n is the partial map $C_n : \mathbb{N} \times \{1,\dots,n\} \to S$ with $C_n(0,1) = \mathsf{G}$, $C_n(0,i) = \mathsf{L}$ for $i \geq 2$, and $C_n(t+1,i) = \delta(C_n(t,i-1), C_n(t,i), C_n(t,i+1))$ if these three arguments are defined and none is F, $C_n(t+1,i)$ undefined otherwise, where $C_n(t,0) = C_n(t,n+1) = *$. The rule *synchronizes* it at time T if $C_n(T,i) = \mathsf{F}$ for $1 \leq i \leq n$ and $C_n(t,i) \in W$ for $t < T$ and $1 \leq i \leq n$. A *minimal-time solution* synchronizes the line of length n at time $2n-2$ for every $n \geq 2$; it has k states if $|S| = k$. The *half-line* $C_\infty : \mathbb{N} \times \{1,2,\dots\} \to S$ is defined in the same way with a border on the left only.

Lemma 2.1. *For every rule, $C_n(t,i) = C_\infty(t,i) = \mathsf{L}$ whenever $t < i-1$ (and $i \leq n$ for C_n).*

Proof. Induction on t; for $t = 0$ this is the initial condition. For $t \geq 1$ the predecessors $(t-1,i')$, $i' \in \{i-1,i,i+1\}$, satisfy $t-1 < i'-1$, so they are L or the right border, and (1) gives L. □

Lemma 2.2 (agreement)**.** *For every rule, $n \geq 2$, $1 \leq i \leq n$ and $t+i \leq 2n-2$, the values $C_n(t,i)$ and $C_\infty(t,i)$ are both undefined or both defined and equal.*

Proof. Induction on t, the case $t = 0$ being clear. If $i = n$, then $t \leq n-2 < i-1$ and both are L by Lemma 2.1. If $i < n$, the predecessors $(t-1,j)$, $j \in \{i-1,i,i+1\}$, satisfy $(t-1)+j \leq 2n-2$ and $j \leq n$, so they are both undefined or equal by induction (for $j = 0$ both are the border), and the recursion gives the claim. □

The border is felt only in the *reflected triangle* $R_n := \{(t,i) : 1 \leq i \leq n,\ t+i \geq 2n-1,\ t \leq 2n-2\}$.

Lemma 2.3 (return chain)**.** *If δ synchronizes the lines of lengths n and $n+1$ at times $2n-2$ and $2n$, then $C_n(2n-1-i,i) \neq C_\infty(2n-1-i,i)$ for $1 \leq i \leq n$, and the* front *$f_i := C_\infty(i-1,i)$ satisfies $f_{n-1} \neq \mathsf{L}$.*

Proof. For $i = 1$, $C_n(2n-2,1) = \mathsf{F}$, whereas $C_\infty(2n-2,1) = C_{n+1}(2n-2,1) \neq \mathsf{F}$ by Lemma 2.2 and since $2n-2 < 2n$. If the claim holds for $i < n$, then the left and middle predecessors of $(2n-1-i,i)$ have coordinate sums $2n-3$ and $2n-2$ and agree in C_n and C_∞ (for $i = 1$ the left one is the border), so the right ones differ: this is the claim for $i+1$. For $i = n$ it reads $\delta(f_{n-1},\mathsf{L},*) \neq \delta(f_{n-1},\mathsf{L},\mathsf{L})$ (by Lemmas 2.1 and 2.2), which is impossible if $f_{n-1} = \mathsf{L}$ by (1). □

If δ is a minimal-time solution and $n \geq t+i+1$, then $C_\infty(t,i) = C_n(t,i)$ with $t < 2n-2$ by Lemma 2.2; hence

$$C_\infty(t,i) \in W \qquad \text{for all } t \geq 0,\ i \geq 1. \tag{2}$$

3. The right end

Fix $n \geq 2$ and use the relative coordinates $\tau := t-(n-1)$, $\kappa := n-i$, in which $R_n = \{(\tau,\kappa) : 0 \leq \kappa \leq \tau \leq n-1\}$, with the right border at $\kappa = -1$ and the left border at $\kappa = n$; write $R_n(\tau,\kappa) := C_n(n-1+\tau, n-\kappa)$. The predecessors of a cell of R_n lie in R_n, on a border, or on one of the anti-diagonals $t+i = 2n-3$, $t+i = 2n-2$, where Lemma 2.2 applies. The *input word* of R_n is the partial word

$$J_n(\kappa) := \big(C_\infty(n-3+\kappa,\ n-\kappa),\ C_\infty(n-2+\kappa,\ n-\kappa)\big), \qquad 0 \leq \kappa \leq n-1,$$

with *lower* and *upper input cells* on $t+i = 2n-3$ and $2n-2$ (for $n = 2$, $\kappa = 0$ the lower cell lies at time -1, is a predecessor of no cell of R_n, and is set to L).

Lemma 3.1 (determinacy)**.** *There is a partial map Ψ, depending on δ only, such that for all $n \geq 2$ and $(\tau, \kappa) \in R_n$ the values $R_n(\tau, \kappa)$ and $\Psi(J_n, n; \tau, \kappa)$ are both undefined or both defined and equal, and $\Psi(J, m; \tau, \kappa)$ depends only on the input cells and borders in the backward dependency cone of (τ, κ).*

Proof. Define $\Psi(J, m; \tau, \kappa)$ for $0 \leq \kappa \leq \tau \leq m-1$ by induction on τ as $\delta(x_{\kappa+1}, x_\kappa, x_{\kappa-1})$, where $x_{\kappa'}$ is given by the first applicable case: $*$ if $\kappa' \in \{-1, m\}$; the lower or upper component of $J(\kappa')$ if $\tau - 1 = \kappa' - 2$ or $\tau - 1 = \kappa' - 1$; otherwise $\Psi(J, m; \tau - 1, \kappa')$ (then $0 \leq \kappa' \leq \tau - 1$); Ψ is undefined where an argument is F or undefined. For $m = n$ these are the left, middle and right predecessors of the cell $(n - 1 + \tau, n - \kappa)$, and on the input cells C_n and C_∞ are both undefined or equal (Lemma 2.2), so induction on τ gives the first claim; unfolding the definition gives the second. □

For $(\tau, \kappa) \in R_n$ let $\mathrm{cone}_n(\tau, \kappa)$ be the set of input cells reached from (τ, κ) by repeatedly passing to predecessors, stopping at input cells and borders.

Lemma 3.2. *For $n \geq 2$, $\mathrm{cone}_n(n-1, 0)$ avoids the left border and consists of the upper input cell at $\kappa' = 0$, both input cells at $1 \leq \kappa' \leq \lfloor n/2 \rfloor$ and, if n is odd, the lower one at $\kappa' = (n+1)/2$.*

Proof. A step changes τ by -1 and κ by at most one, and input cells satisfy $\tau' \in \{\kappa' - 1, \kappa' - 2\}$; a path from $(n-1, 0)$ reaching an input cell at κ' after r steps has $\kappa' \leq r$ and $n - 1 - r \in \{\kappa' - 1, \kappa' - 2\}$, so $2\kappa' \leq n + 1$, and $2\kappa' \leq n$ for upper cells. Conversely, the upper cell at $\kappa' \leq n/2$ is reached by κ' left moves followed by vertical moves, and the lower cell at $1 \leq \kappa' \leq (n+1)/2$ by $\kappa' - 1$ left moves, vertical moves down to $(\kappa' - 1, \kappa' - 1)$ and one more left move, through cells of R_n; the lower cell at $\kappa' = 0$ is a predecessor of no cell of R_n. Every cell of R_n on such a path satisfies $\kappa \leq \tau \leq n - 1 - \kappa$, hence $\kappa \leq (n-1)/2$, so no such cell is adjacent to the left border $\kappa = n$. □

Lemma 3.3 (pumping)**.** *Let $2 \leq n < n'$ and let δ synchronize the lines of lengths n and n' at times $2n - 2$ and $2n' - 2$. Then J_n and $J_{n'}$ differ on $\mathrm{cone}_n(n-1, 0)$.*

Proof. Otherwise, by Lemmas 3.1 and 3.2, the relative cell $(n-1, 0)$ of $R_{n'}$ (it lies in $R_{n'}$, and its cone has the same positions, away from the left border $\kappa = n'$) has the value of the relative cell $(n-1, 0)$ of R_n, namely $C_n(2n-2, n) = \mathsf{F}$. Thus $C_{n'}(n' + n - 2, n') = \mathsf{F}$ although $n' + n - 2 < 2n' - 2$. □

We write $\mathrm{PUMP}(n, n')$ for the conclusion of Lemma 3.3; it is a condition on C_∞ alone. In the frame of the front, $\Phi_t(j) := C_\infty(t, t + 1 - j)$ for $0 \leq j \leq t$ is the cell at *depth* j behind the front, and the recurrence becomes

$$\Phi_{t+1}(j) = \delta\big(\Phi_t(j),\, \Phi_t(j-1),\, \Phi_t(j-2)\big) \qquad (0 \leq j \leq t+1), \tag{3}$$

with $\Phi_t(-1) = \Phi_t(-2) = \mathsf{L}$ and $\Phi_t(t+1) = *$: information only travels to larger depths, and the *depth-row* $t \mapsto \Phi_t(j)$ starts at time j. In these coordinates $J_n(\kappa) = (\Phi_{n-3+\kappa}(2\kappa - 2), \Phi_{n-2+\kappa}(2\kappa - 1))$.

Theorem 3.4 (speed-1/3 barrier)**.** *Let δ be a minimal-time solution. Every depth-row $t \mapsto \Phi_t(j)$ is eventually periodic; let $\theta(j)$ be the least $t_0 \geq j$ such that it is periodic on $[t_0, \infty)$. For every $n \geq 4$ there is a depth $j \leq n$ with $\theta(j) > n + \frac{j}{2} - \frac{5}{2}$. Hence $\limsup_{j \to \infty} \theta(j)/j \geq 3/2$: along infinitely many depth-rows the non-periodic part extends, up to an arbitrarily small angle, to the line $i = t/3$.*

Proof. Row 0 satisfies $\Phi_{t+1}(0) = \delta(\Phi_t(0), \mathsf{L}, \mathsf{L})$ and is eventually periodic; by (3), row j is a finite-state system driven by the eventually periodic rows $j-1, j-2$, hence eventually periodic, with some period P_j. Fix $n \geq 4$ and a common multiple P of $P_0, \ldots, P_{n-1}$. The input cells of the cone at $\kappa = 0$ are quiescent in R_n and R_{n+P} (Lemma 2.1); the others have depth $2\kappa - 2$ (lower) or $2\kappa - 1$ (upper), at most $n - 1$ by Lemma 3.2. If each of them lies in the periodic part of its row, i.e. $n - 3 + \kappa \geq \theta(2\kappa - 2)$ for the lower and $n - 2 + \kappa \geq \theta(2\kappa - 1)$ for the upper ones, then $J_{n+P} = J_n$ on the cone, contradicting Lemma 3.3. So some such cell, of depth $j \leq n$, has $\theta(j) > n - 3 + \kappa \geq n + \frac{j}{2} - \frac{5}{2}$. If $\limsup \theta(j)/j < 3/2$, then $\theta(j) \leq \alpha j + \beta$ for all j with some $\frac{1}{2} \leq \alpha < \frac{3}{2}$ and β, and for n large and all $j \leq n$ we get $(\alpha - \frac{1}{2})j \leq (\alpha - \frac{1}{2})n < n - \beta - \frac{5}{2}$, i.e. $\theta(j) \leq n + \frac{j}{2} - \frac{5}{2}$, a contradiction. Finally, the cell of depth j at time $\theta(j)$ is at position $\theta(j) + 1 - j$, which is at least $\theta(j)/3 + 1$ when $\theta(j) \geq 3j/2$. □

In simulations (up to times 2600, resp. 1400), Mazoyer's solution has $\theta(j) \in \{2j+1, 2j+2\}$ for $1 \le j < 700$, whereas the rule δ_{12} of Section 6 attains the bound, $\theta(j) = \lceil 3j/2 \rceil + 2$ for $10 \le j \le 300$, and satisfies PUMP(n, n') for $4 \le n < n' \le 650$; it is excluded by Corollary 6.4.

The same pigeonhole argument applies along the right border, in the direction of time.

Lemma 3.5 (right band)**.** *Let δ have k states and synchronize the line of length n at time $2n-2$. Let $1 \le d \le \tau_0 \le n-2$ and $\ell \ge 1$, and suppose that the column $\tau \mapsto R_n(\tau, d)$ is ℓ-periodic on $[\tau_0, n-2]$, i.e. $R_n(\tau+\ell, d) = R_n(\tau, d)$ whenever $\tau, \tau+\ell \in [\tau_0, n-2]$. Then $n-1-\tau_0 \le (k-1)^d \ell$.*

Proof. For $d-1 \le \tau \le n-1$ put $\mathbf{b}(\tau) := (R_n(\tau,0), \dots, R_n(\tau, d-1))$. Since $R_n(\tau+1, \kappa) = \delta(R_n(\tau, \kappa+1), R_n(\tau,\kappa), R_n(\tau, \kappa-1))$ with $R_n(\tau,-1) = *$, there is a map Γ, depending on δ and d only, with $\mathbf{b}(\tau+1) = \Gamma(\mathbf{b}(\tau), R_n(\tau, d))$ for $d \le \tau \le n-2$; its arguments lie in $W^d \times W$ because no cell fires before time $2n-2$. At $\tau = n-1$ the band is F^d. Suppose $n-1-\tau_0 > (k-1)^d\ell$. The pairs $(\mathbf{b}(\tau), \tau \bmod \ell)$, $\tau_0 \le \tau \le n-2$, take at most $(k-1)^d\ell$ values, so there are $\tau_0 \le \tau_1 < \tau_2 \le n-2$ with $\mathbf{b}(\tau_1) = \mathbf{b}(\tau_2)$ and $\ell \mid h := \tau_2 - \tau_1$. For $0 \le s \le n-2-\tau_2$ the driving cells $R_n(\tau_1+s, d)$ and $R_n(\tau_2+s, d)$ lie in $[\tau_0, n-2]$ and differ by a multiple of the period, so they are equal, and induction on s gives $\mathbf{b}(\tau_1+s) = \mathbf{b}(\tau_2+s)$ for $0 \le s \le n-1-\tau_2$. For $s = n-1-\tau_2$ this reads $\mathbf{b}(n-1-h) = \mathbf{b}(n-1) = \mathsf{F}^d$, although $n-1-h < n-1$: a contradiction. □

4. The left end

Near the left border the reflected triangle is produced by a band of anti-diagonals behind the reflected wave. Put $v_c(i) := C_n(2n-2+c-i,\, i)$ for $c \ge 0$ and $\max(1,c) \le i \le n$, and for $c = -1$ and $1 \le i \le n-1$: the cell of the *chain* $t+i = 2n-2+c$ at position i. Chains -1 and 0 are the input anti-diagonals and chain 1 is the return chain. The predecessors of $(2n-2+c-i, i)$ lie on the chains $c-2$, $c-1$, c, so

$$v_c(i) = \delta\big(v_{c-2}(i-1),\, v_{c-1}(i),\, v_c(i+1)\big) \qquad (c \ge 1), \tag{4}$$

with $v_{c-2}(0)$ and $v_c(n+1)$ standing for the border. If the line of length n is synchronized at time $2n-2$, then $v_i(i) = \mathsf{F}$ and $v_c(i) \ne \mathsf{F}$ for $c < i$. In the skewed coordinates $u_c(y) := v_c(y + \lfloor c/2 \rfloor)$ this becomes

$$u_c(y) = \delta\big(u_{c-2}(y),\, u_{c-1}(y+\varepsilon_c),\, u_c(y+1)\big), \qquad \varepsilon_c = 1 \text{ for even } c,\ \varepsilon_c = 0 \text{ for odd } c, \tag{5}$$

with $u_0(y) = C_\infty(2n-2-y, y)$ and $u_{-1}(y) = C_\infty(2n-2-y, y-1)$; here $u_c(y)$ is the cell $y + \lfloor c/2 \rfloor$ at time $2n-2+\lceil c/2 \rceil - y$, which is F if $\lceil c/2 \rceil = y$ and not F if $\lceil c/2 \rceil < y$. Each slice $(u_1(y), u_2(y), \dots)$ is thus computed from the slice $y+1$ and the input cells $u_{-1}(y), u_0(y)$. A word w on an integer interval I is *ℓ-periodic* if $w(y+\ell) = w(y)$ whenever $y, y+\ell \in I$.

Lemma 4.1 (left border)**.** *Let δ have k states and synchronize the line of length n at time $2n-2$. Let $s, \ell \ge 1$ and $s \le Y \le n-s$, and suppose that the words $y \mapsto C_\infty(2n-2-y, y)$ $(s \le y \le Y)$ and $y \mapsto C_\infty(2n-3-y, y)$ $(s \le y \le Y-1)$ are ℓ-periodic. Then $Y - s \le (k-1)^{2s}\ell$.*

Proof. For $s \le y \le Y$ let $\mathbf{u}(y) := (u_1(y), \dots, u_{2s}(y))$; these are cells $y + \lfloor c/2 \rfloor \le n$ with $\lceil c/2 \rceil \le s \le y$, so they are defined, and none is F when $y > s$. By (5), $u_c(y)$, $1 \le c \le 2s$, is computed from components of index $< c$ of $\mathbf{u}(y)$, of index $\le c$ of $\mathbf{u}(y+1)$, and from $u_{-1}(y), u_0(y)$; hence $\mathbf{u}(y) = \Gamma(\mathbf{u}(y+1), u_{-1}(y), u_0(y))$ for $s+1 \le y \le Y-1$, with Γ depending on δ only. Unfolding (5) for even c, $u_{2j}(y) = \delta(u_{2j-2}(y), u_{2j-1}(y+1), u_{2j}(y+1))$, shows that $u_{2s}(y) = \Gamma'(\mathbf{u}(y+1), u_0(y))$ for $s \le y \le Y-1$. By hypothesis, u_0 is ℓ-periodic on $[s, Y]$ and u_{-1} on $[s+1, Y]$.

Suppose $Y - s > (k-1)^{2s}\ell$. The pairs $(\mathbf{u}(y), y \bmod \ell)$, $s+1 \le y \le Y$, take at most $(k-1)^{2s}\ell$ values, so there are $s+1 \le y_1 < y_2 \le Y$ with $\mathbf{u}(y_1) = \mathbf{u}(y_2)$ and $\ell \mid h := y_2 - y_1$. By downward induction, $\mathbf{u}(y) = \mathbf{u}(y+h)$ for $s+1 \le y \le y_1$: if it holds for $y+1 \le y_1$, then $\mathbf{u}(y) = \Gamma(\mathbf{u}(y+1), u_{-1}(y), u_0(y)) = \Gamma(\mathbf{u}(y+1+h), u_{-1}(y+h), u_0(y+h)) = \mathbf{u}(y+h)$, as $y+h < y_2 \le Y$. Since $s+h \le Y-1$ and $u_0(s+h) = u_0(s)$,

$$u_{2s}(s+h) = \Gamma'(\mathbf{u}(s+1+h), u_0(s+h)) = \Gamma'(\mathbf{u}(s+1), u_0(s)) = u_{2s}(s) = \mathsf{F},$$

although $\lceil 2s/2 \rceil = s < s+h$, so that $u_{2s}(s+h)$, the cell $2s+h$ at time $2n-2-h$, is not F: a contradiction. □

Theorem 4.2 (left-border barrier)**.** *Let δ be a minimal-time solution. There are no $s \geq 1$, $\alpha > 0$, t_0 and $p, q \geq 1$ such that $C_\infty(t+p, i) = C_\infty(t, i)$ and $C_\infty(t, i+q) = C_\infty(t, i)$ whenever the cells involved lie in the wedge $\Omega := \{(t, i) : \ s \leq i \leq \alpha t, \ t \geq t_0\}$.*

Proof. Suppose otherwise and put $\ell := pq$. If (t, i) and $(t-\ell, i+\ell)$ lie in Ω, so do the cells $(t-\ell, i+jq)$, $0 \leq j \leq p$, and $(t-\ell+jp, i)$, $0 \leq j \leq q$, hence $C_\infty(t-\ell, i+\ell) = C_\infty(t-\ell, i) = C_\infty(t, i)$. Let $Y_n := \lfloor \alpha(2n-3)/(1+\alpha) \rfloor$. For $s \leq y \leq Y_n$ the cells $(2n-2-y, y)$ and $(2n-3-y, y)$ satisfy $y \leq \alpha(2n-3-y)$, so they lie in Ω once $2n-3-Y_n \geq t_0$, i.e. for n large; since a step of ℓ positions along an anti-diagonal is the step from (t, i) to $(t-\ell, i+\ell)$, the words of Lemma 4.1 are ℓ-periodic on $[s, Y_n]$ and $[s, Y_n - 1]$. As $Y_n \geq \frac{2\alpha}{1+\alpha} n - 4$, for n large $Y := \min(Y_n, n-s)$ satisfies $Y - s > (k-1)^{2s}\ell$, contradicting Lemma 4.1. □

The half-line of a rule is *eventually regular* if there are t_1, $P \geq 1$ and, for each residue $r \in \{0, \ldots, P-1\}$, an integer m, words $w_0, \ldots, w_m$, nonempty words $z_1, \ldots, z_m$ over W and integers $a_j, D_j \geq 0$ (all depending on r) such that for all $\nu \geq 0$ the configuration $C_\infty(t, 1) \cdots C_\infty(t, t+1)$ at time $t = t_1 + r + \nu P$ is $w_0 z_1^{a_1+\nu D_1} w_1 \cdots z_m^{a_m+\nu D_m} w_m$: finitely many periodic regions separated by signals of rational speeds.

Corollary 4.3. *The half-line of a minimal-time solution is not eventually regular.*

Proof. Suppose it is. For each residue r some D_j is positive, because the configuration grows by P cells per period; let $j(r)$ be the least such index. The blocks and junctions left of $z_{j(r)}$ have constant lengths, so the block $z_{j(r)}^{a_{j(r)}+\nu D_{j(r)}}$ starts at a fixed cell s_r and its length grows at least like βt, for some $\beta > 0$ and t large. Inside it the state of a cell depends only on its distance to s_r modulo $|z_{j(r)}|$, and is the same at the times t and $t+P$. With $s := \max_r s_r$, $q := \prod_r |z_{j(r)}|$, $\alpha := \beta/2$ and t_0 large, every cell of $\{(t, i) : s \leq i \leq \alpha t, \ t \geq t_0\}$ lies in the block of index $j(r)$ of its residue class, so C_∞ is periodic with periods P in time and q in space on this wedge, contradicting Theorem 4.2. □

5. The first rows after the reflection

For $T \geq 0$ let $E(T) := \{(\tau, \kappa) : 0 \leq \kappa \leq \tau \leq T\}$, the first $T+1$ rows of a reflected triangle. If $n \geq T+2$, then $E(T) \subseteq R_n$ and no cell of $E(T)$ lies in the firing row $\tau = n-1$.

Lemma 5.1 (early rows)**.** *Let δ be a rule, $T \geq 0$, and $n, n' \geq T+2$ such that for $0 \leq \kappa \leq T+1$ the partial values $J_n(\kappa)$ and $J_{n'}(\kappa)$ coincide. Then for every $(\tau, \kappa) \in E(T)$ the values $R_n(\tau, \kappa)$ and $R_{n'}(\tau, \kappa)$ are both undefined or both defined and equal.*

Proof. By Lemma 3.1 we may compare $\Psi(J_n, n; \tau, \kappa)$ and $\Psi(J_{n'}, n'; \tau, \kappa)$. Unfolding the recursion of Lemma 3.1 from $(\tau, \kappa) \in E(T)$ visits only pairs (τ'', κ'') with $0 \leq \kappa'' \leq \tau'' \leq T$ and reads the symbols $x_{\kappa'}$ with $\kappa' \leq \kappa'' + 1 \leq T+1$. The case $\kappa' = m$ would require $m \leq T+1$, which is excluded for $m \in \{n, n'\}$; hence the left border is never read, and the input word is read only at positions $\kappa' \leq T+1$, where J_n and $J_{n'}$ coincide. The two recursions are therefore identical step by step, including the points where an argument is F or undefined. □

Lemma 5.2 (periodicity certificate)**.** *Let δ be a rule, $D \geq 0$, and $V_t := (\Phi_t(0), \ldots, \Phi_t(D))$ for $t \geq D$. There is a map G_D on W^{D+1}, depending on δ and D only, with $V_{t+1} = G_D(V_t)$ whenever $t \geq D$ and $V_t \in W^{D+1}$. If $V_{t_0} = V_{t_0+p}$ for some $t_0 \geq D$ and $p \geq 1$, and $V_t \in W^{D+1}$ for $t_0 \leq t < t_0 + p$, then $V_t \in W^{D+1}$ and $V_{t+p} = V_t$ for all $t \geq t_0$.*

Proof. For $t \geq D$ and $0 \leq j \leq D$ we have $j \leq t$, so the arguments in (3) are $\Phi_t(j)$, $\Phi_t(j-1)$, $\Phi_t(j-2)$ with indices at most D (L for negative indices); none is the border symbol $\Phi_t(t+1) = *$. Hence $G_D(v)_j := \delta(v_j, v_{j-1}, v_{j-2})$, with $v_{-1} = v_{-2} = \mathsf{L}$, does it. For the second statement, suppose that $V_{t'} \in W^{D+1}$ and $V_{t'+p} = V_{t'}$ for $t_0 \leq t' \leq t$ (true for $t = t_0$). Then $V_{t+1+p} = G_D(V_{t+p}) = G_D(V_t) = V_{t+1}$, and $V_{t+1} \in W^{D+1}$, by hypothesis if $t+1 < t_0 + p$ and because $V_{t+1} = V_{t+1-p}$ otherwise; induction on t concludes. □

Corollary 5.3 (finitely many early rows)**.** *Let δ, $T \geq 0$, $D := 2T+1$, $t_0 \geq D$ and $p \geq 1$ satisfy the hypothesis of Lemma 5.2. Then $J_n(\kappa) = J_{n+p}(\kappa) \in W^2$ for all $0 \leq \kappa \leq T+1$ and $n \geq t_0 + 3$. Hence, for every $N_0 \geq T+2$, every R_n with $n \geq N_0$ coincides on $E(T)$, in the sense of Lemma 5.1, with $R_{n'}$ for one of the finitely many lengths $N_0 \leq n' < \max(N_0, t_0+3) + p$.*

Proof. For $1 \leq \kappa \leq T+1$ the components of $J_n(\kappa)$ are depth-row cells of depths $2\kappa - 2, 2\kappa - 1 \leq D$ at the times $n-3+\kappa$ and $n-2+\kappa$, which are at least $n-2 > t_0$; by Lemma 5.2 they lie in W and are unchanged when n is replaced by $n+p$. For $\kappa = 0$ the lower component is L by convention or by Lemma 2.1, and the upper one is $C_\infty(n-2, n) = \mathsf{L}$. If $n \geq \max(N_0, t_0+3) + p$, subtracting a multiple of p gives n' in $[\max(N_0, t_0+3), \max(N_0, t_0+3) + p)$ with $J_{n'} = J_n$ on $[0, T+1]$, and Lemma 5.1 applies since $n, n' \geq T+2$. □

So the condition that no R_n, $n \geq N_0$, is F on $E(T)$ reduces to finitely many lengths, with a certificate found by simulating the half-line. With complete lines, such rows give finite formulas.

Definition 5.4. Let $k \geq 3$, $T \geq 1$, $N_1 \geq T+2$ and $T+1 \leq C \leq N_1$. A k-state rule δ satisfying (1) *satisfies* $\mathcal{G}_k(T, N_1, C)$ if

(G1) δ synchronizes the line of length n at time $2n-2$ for $2 \leq n \leq T+1$;

(G2) for $T+2 \leq n \leq N_1$, $1 \leq i \leq n$ and $t \leq n-1+T$: $C_n(t,i) \in W$;

(G3) for $T+2 \leq n \leq N_1$ and $1 \leq i \leq n$ with $t := 2n-1-i \leq n-1+T$: $C_n(t,i) \neq C_\infty(t,i)$;

(G4) $C_\infty(t,i) \in W$ whenever $t+i \leq 2N_1$;

(G5) for $T+2 \leq n \leq C$: $R_n(\tau,\kappa) \in W$ for all $(\tau,\kappa) \in R_n$ with $\tau+\kappa \leq n-1$ and $\tau < n-1$, $R_n(n-1,0) = \mathsf{F}$, and $R_n(\kappa,\kappa) \neq C_\infty(n-1+\kappa, n-\kappa)$ for $2\kappa \leq n-1$.

We write $\mathcal{G}_k(T, N_1)$ for $\mathcal{G}_k(T, N_1, T+1)$, in which (G5) is void.

The cells of (G5) form the backward dependency cone of the firing cell $(n-1, 0)$; by Lemma 3.1 their values depend only on the input cells of that cone, and by the local definition of C_n they are defined even if a cell outside the cone is F earlier.

Lemma 5.5. *Every k-state minimal-time solution satisfies $\mathcal{G}_k(T, N_1, C)$ for all T, N_1 and C as in Definition 5.4.*

Proof. (G1) holds by definition. For (G2), $n \geq T+2$ gives $n-1+T < 2n-2$, and every cell of the line of length n is in W before time $2n-2$. (G3) is Lemma 2.3, which applies because the lines of lengths n and $n+1$ are synchronized at times $2n-2$ and $2n$. (G4) is (2). For (G5), the cells of R_n with $\tau < n-1$ are in W, $R_n(n-1,0) = C_n(2n-2, n) = \mathsf{F}$, and the return chain differs from C_∞ by Lemma 2.3. □

6. Applications to four and five states

Encodings and evidence. The CNF formula $\mathrm{MT}_k(\mathcal{N}; M, N')$ has one-hot variables for the entries of a k-state rule table, the cells of C_∞ with $t+i \leq M$ and the cells of R_n, $n \in \mathcal{N}$. Its clauses state the recursion, that no half-line cell is F, that the line of length $n \in \mathcal{N}$ is synchronized at time $2n-2$, the return chains of Lemma 2.3 for $n, n+1 \in \mathcal{N}$, and $\mathrm{PUMP}(n, n')$ for $4 \leq n < n' \leq N'$ (none in $\mathrm{MT}_k(\mathcal{N}; M)$). The formula $\mathcal{G}_k(T, N_1, C)$ is encoded likewise with $M = 2N_1$; its regions are closed under predecessors, so the encoding is equivalent to Definition 5.4; for a fixed germ (defined below), (G2), (G3) can be imposed on all $n \geq N_0$ through the representatives of Corollary 5.3. Unsatisfiability is certified by LRAT proofs [15] from CaDiCaL [16], or from Kissat [17] via `drat-trim`, checked by `lrat-check` [18]; the generators, the enumeration and the simulations are trusted programs. As tests, Mazoyer's solution [19] satisfies $\mathrm{MT}_6(\{3, \ldots, 12\}; 78, 40)$ and, with its table fixed, (G2), (G3) for $T = 8$ and all $n \geq 15$; with the table of δ_{14} below fixed, the formulas are satisfiable for the lines $2, \ldots, 14$ and unsatisfiable with the line 15, or with the early rows ($T = 8$, not $T = 6$) of the lines $15, \ldots, 30$, as simulation predicts; every solver model was re-simulated by an independent checker.

Theorem 6.1. *No four-state rule satisfying* (1)

(i) synchronizes all lines of lengths $2, \dots, 9$ *in minimal time;*

(ii) satisfies $\mathcal{G}_4(6, 10)$*: the lines* $2, \dots, 7$ *complete and, for the lines* $8, 9, 10$*, the first seven rows and the return chains;*

(iii) satisfies $\mathcal{G}_4(4, 10, 10)$*: the lines* $2, \dots, 5$ *complete and, for the lines* $6, \dots, 10$*, the first five rows, the return chains and the right ends.*

In particular, no four-state minimal-time solution exists.

Proof. (i) Such a rule yields a model of $\mathrm{MT}_4(\{2, \dots, 9\}; 16)$: a half-line cell with $t + i \leq 16$ is quiescent if $i \geq 10$ (Lemma 2.1) and equals a cell of the line of length 9 at a time $t < 16$ if $i \leq 9$ (Lemma 2.2), so it is in W, and the return chains hold by Lemma 2.3 for $2 \leq n \leq 8$. CaDiCaL refutes this formula (753 variables, 17,605 clauses) in 51 s with an LRAT proof of 177 MB, which `lrat-check` verifies in 5.7 s. (ii) The encoding of $\mathcal{G}_4(6, 10)$ has 919 variables and 22,960 clauses; CaDiCaL refutes it in 41 s with an LRAT proof of 167 MB, verified in 5 s. (iii) The encoding of $\mathcal{G}_4(4, 10, 10)$ has 874 variables and 20,461 clauses; CaDiCaL refutes it in 36 s with an LRAT proof of 173 MB, verified in 4 s. The SHA-256 digests of the three formulas begin with `d18d6548`, `0e0ba286` and `124f0954`. The last claim follows from (i), and also from (ii) or (iii) with Lemma 5.5. Part (i) certifies the bound observed by Yunès [8]. □

The bounds are sharp in the following sense. An exhaustive enumeration finds 27 partial rules (rules restricted to the transitions they use) for the lengths $2, \dots, 8$, as did Yunès [8] under the extra condition $\delta(*, \mathsf{L}, \mathsf{L}) = \mathsf{L}$, vacuous here as none of them uses $(*, \mathsf{L}, \mathsf{L})$. The encodings of $\mathcal{G}_4(5, 30)$, $\mathcal{G}_4(6, 9)$, $\mathcal{G}_4(4, 30, 9)$, $\mathcal{G}_4(3, 30, 12)$, of $\mathcal{G}_4(6, 10)$ without (G3), and of (G1) for $T = 4$ with (G5) for $6 \leq n \leq 12$ have models. The five-state rules below are the files `delta14*.txt`, `delta13.txt` and `uniformA_2-12.txt` (δ_{12}) in `results/` of the repository.

Proposition 6.2. *Each of the five-state rules* δ_{14}, δ_{14}^{b}, δ_{14}^{c}, δ_{14}^{d}, δ_{14}^{e} *synchronizes the line of length* n *at time* $2n - 2$ *for* $2 \leq n \leq 14$ *and for no* $n \geq 15$*: for* $n \geq 15$*, some cell of the line of length* n *is* F *at a relative time* $\tau \leq \min(T, n - 2)$*, where* $T = 10, 14, 22, 12, 14$*, respectively.*

Proof. The lines of lengths $2, \dots, 60$ are simulated; for $15 \leq n \leq 60$ some cell is F at a relative time $\tau \leq n - 2$, which proves the claim for $15 \leq n < T + 3$. The half-line has no cell in state F at the times $t < 3000$, so by Lemma 2.2 a cell of a line of length $n \leq 1500$ that is F at a time $t \leq 2n - 2$ lies in R_n. Simulating the half-line gives t_0 and p satisfying the hypothesis of Lemma 5.2 for $D = 2T + 4 \geq 2T + 1$, namely $(t_0, p) = (51, 8)$, $(66, 3)$, $(257, 192)$, $(95, 48)$, $(56, 2)$. For every n with $\max(15, T + 3) \leq n < t_0 + 3 + p \leq 452$, direct simulation shows that the line of length n, or of a length n' in this range with $J_{n'} = J_n$ on $[0, T + 1]$, has a cell in state F at a relative time $\tau \leq T$, hence in $E(T)$ (one length is simulated per class of a partition finer than equality of J_n on $[0, T + 1]$; with the lengths below $T + 3$ these are 35, 39, 413, 116, 28 lengths, the largest being 49, 53, 427, 130, 42). By Lemma 5.1 and Corollary 5.3 with $N_0 = \max(15, T + 3)$, every R_n with $n \geq N_0$ is F at a cell of $E(T)$, where $\tau \leq T \leq n - 3$. □

For $3 \leq t < 1500$ the half-line of δ_{14} takes the values L, G at cell 1, L, A at cell 2 and L, B at the cells $i \geq 3$, where δ_{14} restricted to $\{\mathsf{L}, \mathsf{B}\}^3$ is the elementary rule 22 [20]; in the simulated ranges ($\mathrm{PUMP}(n, n')$ for $4 \leq n < n' \leq 800$; $n \leq 5000$, $s \leq 3$, $\ell \leq 24$ in Lemma 4.1) neither barrier excludes it. The rules δ_{12} and δ_{13} synchronize the lines of lengths $2, \dots, 12$, resp. $2, \dots, 13$, and no other line of length at most 60. By simulation, each of these seven rules has a line of some length $14 \leq n \leq 19$ with a cell in state F at a relative time $\tau \leq 9$, so none of them satisfies $\mathcal{G}_5(9, N_1)$ for $N_1 \geq 19$.

Half-lines of low complexity. The *germ* below M of a rule is its restriction to the neighbourhoods of the cells (t, i), $t \geq 1$, $t + i \leq M$, of its half-line, and its *complexity* $c_M(\delta)$ is the number of these neighbourhoods other than $(\mathsf{L}, \mathsf{L}, \mathsf{L})$: $c_{78} = 57$ for Mazoyer's rule, 22 for δ_{12} and 16 for δ_{14}.

Proposition 6.3. *Every five-state minimal-time solution* δ *satisfies* $c_{78}(\delta) \geq 15$.

Proof. Suppose $c_{78}(\delta) \le 14$; exchanging A and B if necessary, A occurs before B (if at all) in the order of anti-diagonals and, within an anti-diagonal, of time. By (2) and Lemmas 2.3 and 3.3, the germ of δ below 78 produces no F, never produces L at the front, and satisfies PUMP(n, n') for $4 \le n < n' \le 40$. A depth-first search that chooses the value of each neighbourhood at its first occurrence enumerates all such germs ($1.0 \cdot 10^8$ nodes): there are 244, and an independent re-implementation finds the same 244. For 230 of them, $\mathrm{MT}_5(\{2, \dots, 10\}; 78, 39)$ (without symmetry breaking) with the germ fixed is unsatisfiable: one formula with a selector variable per germ, clauses that make each selector fix its germ and a clause requiring some selector is refuted by CaDiCaL in 4 s with an LRAT proof of 11 MB. Each of the other 14 germs determines C_∞ up to the anti-diagonal 3300, and there Lemma 4.1 applies with $s = 1$: for some $n \le 163$ and $\ell \le 5$, the upper input anti-diagonal of the line of length n is ℓ-periodic on $[1, Y]$ and the lower one on $[1, Y-1]$ with $Y - 1 > 16\ell$ (e.g. $n = 28$, $\ell = 1$, $Y = 19$), as an independent simulation confirms. Hence no rule containing one of the 244 germs is a minimal-time solution. □

The rules δ_{12} and δ_{13} have the same half-line for $t + i \le 78$, where they use the 22 transitions

```
*GL>A GLL>G *AG>B AGL>L *BL>A BLG>L LGL>B *AL>G ALB>A LBG>L BGL>G
*GA>A GAL>L ALG>B LGG>G GGL>G BGG>L GGG>G ALL>A LLG>L ALA>A LAL>L
```

(`xyz>d` means $\delta(x, y, z) = d$) besides $\delta(\mathsf{L}, \mathsf{L}, \mathsf{L}) = \mathsf{L}$: a front, a wake of G's and, behind a boundary of speed 1/3, a region $\mathsf{A}\,\mathsf{L}\,\mathsf{A}\,\mathsf{L}\cdots$ periodic in time and space.

Corollary 6.4. *No five-state rule with the* 22 *transitions above synchronizes the line of length* 518 *in minimal time.*

Proof. With these transitions and $\delta(\mathsf{L}, \mathsf{L}, \mathsf{L}) = \mathsf{L}$ only, no other neighbourhood occurs on the cells with $t + i \le 1034$, so every such rule has the same diagram C_∞ there. On $t + i = 1034$ the cells $2, \dots, 259$ are in state A, and on $t + i = 1033$ the cells $2, \dots, 258$ are in state L. Lemma 4.1 with $k = 5$, $n = 518$, $s = 2$, $\ell = 1$, $Y = 259$ would give $257 \le 4^4 = 256$. □

Balzer's conditions. Balzer [5, p. 37] reported that no five-state minimal-time solution satisfies the four conditions (B1) $\delta(x, \mathsf{G}, y) = \mathsf{G}$ unless $x, y \in \{\mathsf{G}, *\}$; (B2) $\delta(I(z), I(y), I(x)) = I(\delta(x, y, z))$ for an involution I of the states with $I(*) = *$; (B3) $\delta(\mathsf{G}, \mathsf{G}, \mathsf{G}) = \delta(*, \mathsf{G}, \mathsf{G}) = \delta(\mathsf{G}, \mathsf{G}, *) = \mathsf{F}$ and no other neighbourhood gives F; (B4) $\delta(\mathsf{G}, v, \mathsf{G}) = \mathsf{G}$ for all $v \in W \setminus \{\mathsf{G}\}$. In the *weak reading* [8], (B1) is required only for working states x, y and (B3) without its last clause. We take $I(\mathsf{F}) = \mathsf{F}$; then $I(\mathsf{G}) = \mathsf{G}$, since otherwise $g := I(\mathsf{G}) \in W \setminus \{\mathsf{G}\}$ and (B2), (B4) give $\delta(g, \mathsf{G}, g) = I(\delta(\mathsf{G}, g, \mathsf{G})) = I(\mathsf{G}) = g$, whereas (B1) gives $\delta(g, \mathsf{G}, g) = \mathsf{G}$. So I is one of the four involutions of $\{\mathsf{L}, \mathsf{A}, \mathsf{B}\}$, and (B1)–(B4) become unit and binary clauses.

Proposition 6.5. *No five-state rule satisfying (B1)–(B4) in the weak reading, with* $I(\mathsf{F}) = \mathsf{F}$*, synchronizes all lines of lengths* $2, \dots, 12$ *in minimal time; a fortiori none satisfying Balzer's form does. The lengths up to* 11 *do not suffice to prove this.*

Proof. For each involution, the clauses of (B1)–(B4) are added to $\mathrm{MT}_5(\{2, \dots, N\}; 2N-2)$. For $I = (\mathsf{A}\,\mathsf{B})$ and $N = 10$, and for $I = (\mathsf{L}\,\mathsf{A})$ and $I = (\mathsf{L}\,\mathsf{B})$ and $N = 9$, CaDiCaL refutes the formula with LRAT proofs checked in 7.1, 3.5 and 3.3 s. For I the identity and $N = 12$, Kissat refutes it in 15 minutes; drat-trim turns its DRAT proof into an LRAT proof of 3.1 GB, which `lrat-check` verifies in 66 s. The rule `results/balzer_sym11.txt` satisfies (B1)–(B4) in the strong reading with I the identity and synchronizes every line of length at most 11 (direct simulation). □

7. Concluding remarks

The five-state rules that come closest to minimal time are irregular and fail in the first rows after the reflection. For the germ of δ_{14}^{d}, (G1)–(G3) with $T = 10$ and all $n \ge 12$ have a model, and so do (G1)–(G3) with $T = 7$ and all $n \ge 9$ together with (G5) for $9 \le n \le 20$; in the latter, the lines of lengths 9, $11, \dots, 17$ and 19 have both end cells, but not all cells, in state F at time $2n - 2$ and none before. Excluding such rules

requires conditions on the middle cells, whose dependency cones cover most of R_n. Open questions: (1) Does a five-state minimal-time solution exist? (2) Is there a condition on the middle of the reflected triangles as economical as Lemma 5.5? (3) Can the bound of Lemma 4.1 be made polynomial in s? (4) Do Balzer's conditions exclude six states?

Declarations

CRediT. Kia Keng Giam: Conceptualization, Methodology, Formal analysis, Investigation, Validation, Writing – review & editing. *Competing interests and funding.* None. *Data availability.* Programs, rule tables, logs, SHA-256 digests and two scripts that regenerate the certificates and check every computational statement are in the supplementary material and at `https://github.com/giamkiakeng/proj1` (directory `fssp5/note2/supplementary`). *Use of generative AI.* The author used Claude (Anthropic) to develop and check the arguments, to write and run the programs and experiments, and to draft and proofread the text. The author reviewed and edited the content and takes full responsibility for it.

References

[1] E. F. Moore, The firing squad synchronization problem, in: E. F. Moore (Ed.), Sequential Machines: Selected Papers, Addison-Wesley, Reading, MA, 1964, pp. 213–214.

[2] E. Goto, A minimal time solution of the firing squad problem, Dittoed course notes for Applied Mathematics 298, Harvard University, pp. 52–59 (1962).

[3] A. Waksman, An optimum solution to the firing squad synchronization problem, Information and Control 9 (1) (1966) 66–78.

[4] T. Correa, B. Gustavo, L. Lemos, A. Settle, An overview of recent solutions to and lower bounds for the firing synchronization problem, arXiv:1701.01045 (2017).

[5] R. Balzer, An 8-state minimal time solution to the firing squad synchronization problem, Information and Control 10 (1) (1967) 22–42.

[6] H.-D. Gerken, Über Synchronisationsprobleme bei Zellularautomaten, Diplomarbeit, Institut für Theoretische Informatik, Technische Universität Braunschweig (1987).

[7] J. Mazoyer, A six-state minimal time solution to the firing squad synchronization problem, Theoretical Computer Science 50 (2) (1987) 183–238.

[8] J.-B. Yunès, Synchronisation et automates cellulaires: la ligne de fusiliers, Ph.D. thesis, Université Paris 7, available from HAL, `tel-00139049` (1993).

[9] P. Sanders, Massively parallel search for transition-tables of polyautomata, in: Parcella '94: Proceedings of the VI International Workshop on Parallel Processing by Cellular Automata and Arrays, Akademie Verlag, Berlin, 1994, pp. 99–108.

[10] J. Mazoyer, On optimal solutions to the firing squad synchronization problem, Theoretical Computer Science 168 (2) (1996) 367–404.

[11] K. Yamashita, Y. Nishitani, S. Hirose, S. Okawa, N. Osato, The firing squad synchronization problem with sub-generals, Information Processing Letters 114 (1–2) (2014) 60–65.

[12] K. Kobayashi, Nonexistence of minimal-time solutions for some variations of the firing squad synchronization problem having simple geometric configurations, arXiv:1909.10125 (2019).

[13] J. Mazoyer, V. Terrier, Signals in one dimensional cellular automata, Research Report 94-50, Laboratoire de l'Informatique du Parallélisme, ENS Lyon, available from HAL, `hal-02101868` (1994).

[14] J. Mazoyer, V. Terrier, Signals in one-dimensional cellular automata, Theoretical Computer Science 217 (1) (1999) 53–80.

[15] L. Cruz-Filipe, M. J. H. Heule, W. A. Hunt, Jr., M. Kaufmann, P. Schneider-Kamp, Efficient certified RAT verification, in: Automated Deduction – CADE 26, Vol. 10395 of Lecture Notes in Computer Science, Springer, 2017, pp. 220–236.

[16] A. Biere, T. Faller, K. Fazekas, M. Fleury, N. Froleyks, F. Pollitt, CaDiCaL 2.0, in: Computer Aided Verification (CAV 2024), Vol. 14681 of Lecture Notes in Computer Science, Springer, 2024, pp. 133–152.

[17] A. Biere, K. Fazekas, M. Fleury, M. Heisinger, CaDiCaL, Kissat, Paracooba, Plingeling and Treengeling entering the SAT Competition 2020, in: Proceedings of SAT Competition 2020: Solver and Benchmark Descriptions, Vol. B-2020-1 of Department of Computer Science Report Series B, University of Helsinki, 2020, pp. 51–53.

[18] N. Wetzler, M. J. H. Heule, W. A. Hunt, Jr., DRAT-trim: efficient checking and trimming using expressive clausal proofs, in: Theory and Applications of Satisfiability Testing (SAT 2014), Vol. 8561 of Lecture Notes in Computer Science, Springer, 2014, pp. 422–429.

[19] J. Duprat, Firing Squad Synchronization Problem, Coq user contribution `FiringSquad` (formal proof of the correctness of Mazoyer's six-state solution), LIP, ENS Lyon; archived at `https://github.com/rocq-archive/firing-squad` (1997).

[20] S. Wolfram, Statistical mechanics of cellular automata, Reviews of Modern Physics 55 (3) (1983) 601–644. `doi:10.1103/RevModPhys.55.601`.